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DATA ANALYTICS COURSE

INTRODUCTION

The data analytics course teaches you how to collect, analyze, and interpret data, helping you make informed decisions and solve problems in different industries. Edoxi's Data Analytics course equips you with the skills to make data-driven decisions by enhancing your expertise in Data Analytics using PoweBI, Statistics, Python, and SQL.

COURSE OVERVIEW

Edoxi Training Institute offers Data Analytics training from our industry-experienced and certified trainers. This comprehensive Data Analytics course equips participants with the essential skills and knowledge to extract valuable insights from data. The course covers a wide range of topics, from Python programming and database management to statistical analysis, data visualization, and machine learning. Through hands-on projects and real-world case studies, participants will gain practical experience in applying data analytics techniques to solve business problems and make data-driven decisions.

WHO CAN ATTEND?

Individuals who can attend the training program include:

- Data Analysts.
- Business Analysts.
- Professionals seeking to enhance their data analysis skills.

COURSE OUTCOME

Upon completion of the program, you will be able to:

- Explore Python programming, covering grammar, data structures, and control flow.
- Use OOP and important libraries such as NumPy, Pandas, and Matplotlib for data manipulation, analysis, and visualization.
- Design, implement, and manage relational databases using MySQL, and build applications that interact with databases.
- Develop insightful and interactive dashboards using Power BI.
- Understand core statistical concepts and draw meaningful insights from data.
- Explore the complete data science workflow, from data acquisition and cleaning to machine learning algorithms and model performance.
- Apply your skills through hands-on projects involving data analysis, database management, interactive dashboards, and predictive modeling.
- Effectively present and share data results using Python, MySQL, and Power BI-generated visuals and reports.

DATA ANALYTICS COURSE OUTLINE

ADVANCED MS EXCEL

MODULE :1

- Recall of Basic Excel.
- Overview of the Basics of Excel.
- Cell Reference (Relative, Absolute, Mixed).
- Circular and 3D cell reference.
- Custom Number Formatting.

MODULE :2

- Date and time functions.
- Text functions.
- Database functions.
- Writing conditional expression (using IF).
- Using logical functions (AND, OR, NOT).
- IF AND OR Function using together.
- Power Functions (CountIF, CountIFS, SumIF, SumIFS) Sumproduct.

MODULE :3

- Using lookup and reference functions (VLOOKUP, HLOOKUP, MATCH, INDEX).
- Vlookup with Exact Match, Approximate Match.
- Nested Vlookup with Exact Match.
- Vlookup with Tables, Dynamic Ranges.
- Nested Vlookup with Exact Match.
- Using Vlookup to consolidate Data from Multiple sheets.
- Xlookup Function with some smart options.

MODULE :4

- Advanced-Data Validation.
  - Specifying a valid rage of values for a cell.
  - Specifying a list of valid values for a cell.
  - Specifying custom validations based on formula for a cell.
  - Making Dynamic List.
  - Creating Dependent Drop-Down List.

- Advanced Conditional Formatting
  - Using auto-formatting option for worksheets.
  - Using conditional formatting option for rows, columns and cells.
  - Wild card Formatting.

- Working with Templates
  - Designing the structure of a template.
  - Using templates for standardization of worksheets.
  - Sorting and filtering Data.

- Sorting Tables
  - Using multiple-level sorting.
  - Using custom sorting.
  - Filtering data for selected view (AutoFilter).
  - Using advanced filter options.

MODULE :5

- Macros
  - Relative & Absolute Macros.
  - Editing Macro's.
  - Creating Automation Button.

- Scenario
  - What-If Analysis
  - Goal Seek , Solver
  - Data Tables
  - Scenario Manager

- Important Features of Excel
  - Flash Fill, Sparklines, Remove Duplicate
  - Text to Column, Freeze Pane, Split
  - Protect File, Sheet, Workbook,
  - Protect Range

MODULE :6

- Recall
- Project on Excel

MICROSOFT POWER BI

MODULE 1: INTRODUCTION TO SELF-SERVICE BI SOLUTIONS

- Introduction to Business Intelligence.
- Introduction to Data Analysis.
- Introduction to Data Visualization.
- Overview and Considerations for Self-Service BI.
- Microsoft Tools for Self-Service BI.

MODULE 2: INTRODUCING POWER BI

- MODULE 2: INTRODUCING POWER BI.
- Explore the rationale of Power BI.
- Define the components of Power BI Service.
- Connecting to conventional Power BI Data sources.
- Create the first Power BI Report using non-Excel data.
- Introduction to advanced data sets (WEB, API, JSON, ODATA, etc.).

MODULE 3: POWER BI DATA

- Using Excel as a Data Source for Power BI.
- Exploring the Power BI Data Model. Using Online/Live
- Databases as a Data Source for Power BI.
- Importing Excel files into Power BI.
- Viewing Reports from Excel Files.
- Exploring smart techniques to get data into Power BI from non-conventional data sets.

MODULE 4: SHAPING AND COMBINING DATA

- Power BI Desktop Queries.
- Power BI Visualization - Tables and Maps.
- Using Geographical Data in Visualizations.
- Shape and Combine Power BI Data - Mashup.
- Perform a range of query editing tasks in Power BI.
- Shape data, using formatting and transformations.
- Combine data from tables in your dataset.

MODULE 5: MODELLING DATA

- Defining the need for data Relationships and understanding cardinality.
- Introduction to DAX Queries.
- DAX Concepts - Row context & Filter context.
- Calculations and Measures.
- Create Relationships and describe relationships between data tables.
- Understanding logical Calculations.
- Understand the DAX syntax and use DAX functions to enhance your datasets Essential DAX functions - CALCULATE, FILTER, RELATEDTABLE.
- Working on Dates - developing date dimensions using.
- Advanced M Query. O Create calculated columns, calculated tables, & measures.
- Important DAX Functions (Time Intelligence and Iterators).
- NOW, TODAY, DATE, DATEADD, DATEDIFF, TOTALMTD...QTD, YTD, SAMEPERIODLASTYEAR.

MODULE 6: INTERACTIVE DATA VISUALIZATION

- MODULE 6: INTERACTIVE DATA VISUALIZATION.
- Creating Power BI Reports.
- Creating a Power BI Dashboard.
- Use Power BI Desktop to create interactive data visualizations.
- Manage a Power BI solution using Custom Visuals.
- Import custom visuals into Power BI for Power BI reports.
- Advanced visualisation techniques - synchronised filters and bookmarks.
- Advanced Power BI features - Filtering, Hierarchy, Drilldown, Tooltips.

MODULE 7: DIRECT CONNECTIVITY

- Advantages of maintaining Cloud Data.
- Publishing Power BI data to the Cloud.
- Connecting to Analysis Services.
- Direct Connections to Power BI.
- Use Power BI with SQL Server Analysis Services data.
- Overview of online data sources ingestion (e.g., Google Analytics).
- Use Analysis Services models running in multidimensional mode.

MODULE 8:	POWER BI MOBILE & WEB SERVICES

- Using the Power BI Mobile App.
- Develop Mobile-compatible responsive reports and dashboards.
- Create dashboards and reports for mobile devices.
- Use Power BI Embedded to add visualizations and reports to web or mobile applications.
- Exploring Power BI Cloud services - Quick Insights, Usage Metrics, Report Scheduling.
- Developing reports using Power BI NLP features.
- Scheduling for auto-refresh data.

1. Python Fundamentals (2 hours)

- Introduction and Setup.
    - Overview of Python, IDEs, and environment setup.
- Basic Syntax and Variables.
    - Variable assignments and basic arithmetic operations.
- Numbers and Strings.
    - Numbers: Types of numbers and basic arithmetic.
    - Strings: Creating a string and variable assignments.
        - String indexing & slicing.
        - String concatenation & repetition.
        - Basic built-in string methods.

2. Data Structures in Python (4 hours)

- Lists.
    - Creation, indexing, slicing, and common list methods.
- Tuple.
    - Creation, indexing, slicing, and common tuple methods.
- Dictionaries.
    - Key-value pairs, accessing and modifying data.
- Sets & Booleans.
    - Set operations, uniqueness; Boolean values and expressions.

3. Python Programming Constructs (2 hours)

- Conditional Statements.
    - if, Elif, else statements.
- Loops & Iterative Statements.
    - for loops and while loops.
- Comprehensions.
    - List, dictionary, and set comprehensions.
- Exception Handling.
    - Try/except blocks and basic error management.

4. Modules, Packages, and File I/O (2 hours)

- Modules and Packages.
    - Importing and organizing code into modules/packages.
- File I/O Operations.
    - Reading from and writing to files.
- Modular Programming in Python.
    - Best practices for structuring larger projects.

5. Functions (2 hours)

- Introduction to Functions.
    - What is a function and why use functions?
- Function Definition (def statement).
    - Writing and calling functions.
- Function Arguments & Return Values.
    - Positional and keyword arguments; handling return data.
- Lambda Functions.
    - Anonymous functions and their common use cases.

6. Object-Oriented Programming (OOP) in Python (2 hours)

- OOP Fundamentals (Objects & Classes).
    - Creating and instantiating classes and objects.
- Attributes and Methods.
    - Instance attributes, methods, and their usage.
- Instance Methods vs. Class Methods vs. Static Methods.
    - Understanding differences, use cases, and examples.

7. NumPy (2 hours)

- Overview of NumPy and Array Creation.
    - Creating arrays using functions such as:
        - np.array
        - np.zeros
        - np.ones
        - np.arange
- Array Attributes.
    - Examining key properties: shape, size, dimensions (ndim), data type (dtype), and item size.
- Indexing and Slicing.
    - Accessing individual elements and slices of arrays.
- Element-wise Operations and Array Math.
    - Performing arithmetic operations on arrays.
    - Utilizing vectorized operations and universal functions (ufuncs).
- Broadcasting.
    - Understanding and applying broadcasting rules to perform operations on arrays with different shapes.
- Array Reshaping and Manipulation.
    - Reshaping arrays using functions like reshape, flatten, and transpose.
- Random Number Generation.
    - Using NumPy's random module to generate arrays of random numbers.

8. Exploratory Data Analysis (EDA) with Pandas (4 hours)

- Introduction to Pandas.
    - Overview of the Pandas library.
- Core Data Structures.
    - Series:
        - Creating a Series.
        - Indexing and slicing in Series.
        - Basic operations on Series.
    - DataFrame:
        - Creating DataFrames from various sources. (lists, dictionaries, arrays).
        - Understanding DataFrame structure and attributes.

          (index, columns, dtypes).

- Data Input/Output Operations.
    - Reading data from different file formats (e.g., CSV, Excel, JSON).
    - Writing DataFrames to files.
- Data Exploration and Inspection.
    - Viewing data using methods like head( ), tail( ), and sample( ).
    - Getting DataFrame summaries with info( ), describe( ), and checking data types.
- Data Cleaning and Preparation.
    - Handling missing values using functions like isnull( ), dropna( ), and fillna( ).
    - Removing duplicates.
- Data Manipulation and Transformation.
    - Selecting, filtering, and subsetting data.
- Merging and Joining Data.
    - Combining DataFrames through merging, joining, and concatenation.
- Grouping and Aggregation.
    - Grouping data using the group by method.

9. Data Visualization Techniques with Matplotlib and Seaborn (4 hours)

- Introduction to Data Visualization.
    - Importance of visualizing data in the analysis process.
    - Overview of Matplotlib and Seaborn libraries.
- Visualization with Matplotlib.
    - Basic plotting techniques:
        - Line plots, bar charts, histograms, scatter plots.
    - Customizing plots:
        - Adding titles, axis labels, legends, and annotations.

Visualization with Seaborn.

Enhancing visualizations with Seaborn's high-level interface.

Creating statistical plots:

Box plots, violin plots, swarm plots, and pair plots.

Visualizing relationships:

Heatmaps for correlation matrices

Statistics for Data Analysts (2 hours)

- Core Statistical Concepts.
    - Measures of central tendency & dispersion, correlation and covariance.
- Basic Probability & Hypothesis Testing.
    - Overview of probability distributions and an introduction to hypothesis tests.

9. Data Visualization Techniques with Matplotlib and Seaborn (4 hours)

- Introduction to Data Visualization.
  - Importance of visualizing data in the analysis process.
  - Overview of Matplotlib and Seaborn libraries.
- Visualization with Matplotlib.
  - Basic plotting techniques:
    - Line plots, bar charts, histograms, scatter plots.
  - Customizing plots:
    - Adding titles, axis labels, legends, and annotations.
- Visualization with Seaborn.
- Enhancing visualizations with Seaborn's high-level interface.
  - Creating statistical plots:
    - Box plots, violin plots, swarm plots, and pair plots.
- Visualizing relationships:
    - Heatmaps for correlation matrices.

MySQL for Data Science (8 hours)

1. Introduction to MySQL in Data Analytics (30 Mins)

- Setting Up and Connecting to MySQL (30 Mins)
  - Installation and Configuration:
  - Installing MySQL server and client tools
  - Configuring MySQL on local machines

2. Core SQL Fundamentals (2 hours)

- Basic SQL Syntax and Queries:
  - Structure of SQL queries (SELECT, FROM, WHERE).
  - Using aliases for tables and columns.
- Data Retrieval:
  - Filtering data with WHERE clauses.
  - Sorting results with ORDER BY.
- CRUD Operations:
  - Inserting data (INSERT).
  - Updating records (UPDATE).
  - Deleting records (DELETE).

3. Data Manipulation and Aggregation (2 hours)

- Aggregation Functions:
  - COUNT, SUM, AVG, MIN, MAX.
  - Using GROUP BY for aggregating data.
  - Filtering groups with HAVING.
- Subqueries and Nested Queries:
  - Writing subqueries in SELECT, WHERE, and FROM clauses.
  - Correlated, non-correlated subqueries.

4. Joining Tables and Data Relationships (2 hours)

- Join Operations:
  - INNER JOIN: combining rows from multiple tables based on common fields.
  - LEFT, RIGHT, and FULL OUTER JOINS: handling unmatched rows.

5. Advanced Query Techniques (2 hours)

- Advanced SQL Queries:
  - Window functions for running totals and moving averages.
  - Common Table Expressions (CTEs) for improved query readability.

ABOUT EDOXI

Edoxi is an ed-tech company established in Dubai in 2018. The company has broadened its offerings to include over 300 sought-after professional and accredited courses such as Cybersecurity, Cloud Computing, Data Science, Data Analytics, Software Programming, Project Management, Office Productivity, soft skills training, etc. With physical locations in Dubai, Qatar, and London, Edoxi has shown a strong commitment to delivering exceptional training to learners worldwide. For over five years, Edoxi has been recognized for providing high-quality, job-focused training to students, professionals, and businesses in various sectors.

Edoxi is approved and accredited by several reputable institutions, such as KHDA, Microsoft, AICERTs, PMI, PECB, British Council, Qualifi, AutoDesk, EC Council, and CompTIA. The company provides Professional certification courses, Customised training programs, Organisational diagnostics, and training consultation services. It prides itself on a team of more than 100 certified trainers who excel in delivering top-notch training across different subjects. Edoxi has garnered satisfaction from over 200 clients globally for its services. Its courses are available through both classroom and online formats, drawing students from over 100 countries.

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